Working Memory, Proficiency, and L2 RC Attachment with Chinese-English Bilinguals

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Introduction. To resolve globally ambiguous relative clauses (RC; e.g., *John called the servant of the actress who was on the balcony*), readers must determine whether the modifier attaches to either of two preceding nouns (HA: *servant;* or LA: *actress*). RC attachment preferences are known to vary across languages and individuals due to a variety of factors. RC disambiguation has also drawn the attention of researchers interested in multilingualism – as a way to shed light on how experience, cognitive resources, and grammatical knowledge interact during sentence processing. Recently, Cotter and Ferreira (2024) report working memory capacity (WMC) effects on RC disambiguation with bilingual Mandarin-English participants *when tested in their L1*, such that WMC was inversely correlated with high attachment (HA) preferences, suggesting that multilingual users recruit WM resources similarly to monolinguals (Swets et al. 2007). However, evidence for WM effects on L2 processing is mixed (Omaki 2005; a.o.). To further probe whether cognitive resources differentially impact bilingual processing in L1 and L2, here we directly extend Cotter and Ferreira and investigate how WMC relates to *L2* attachment preferences.

Materials & Procedure. Our target items were 20 critical sentences taken from the English materials in Cotter and Ferreira (2024), containing a complex NP (*the servant of the actress*) followed by an RC ambiguously modifying either of the two nouns. Each sentence was then followed by a forced-choice comprehension question probing disambiguating preferences (e.g. "*Who was on the balcony*?"). We included 80 filler items, which served as attention checks. Following Cotter and Ferreira (2024), participants completed an assessment of spatial and reading WMC, and they self-assessed their L1 and L2 proficiency through the Language History Questionnaire (LHQ; Li et al. 2020). We additionally included an English Cloze test at the end of the experiment, as a direct measure of L2 proficiency (Montrul 2006). All tasks were in English.

Results (N = 118)**.** We recruited L1-Chinese/L2-English bilinguals through Prolific. We included participants in our analyses only if they completed every task and correctly interpreted at least 80% on the filler items. Descriptive statistics show a slight tendency towards HA (mean:0.57; sd:0.49). However, participants' attachment preference actually appears bi-modally distributed, with high variance (Fig. 1). We also observe a relatively strong correlation between the two WMC spans, and both the self-assessed and the Cloze L2 proficiency scores (Fig. 2). Following Cotter & Ferreira (2024), we thus operationalize WMC as the average of z-transformed reading and spatial spans (AVGM). We then fit logit mixed-effect models to explore the effect of memory and proficiency on HA preference rates. AIC and BIC model comparison revealed that the best fitting model included AVGM, z-scored self-assessed L1 proficiency, and Cloze score and their interactions as fixed effects, and item and participant random effects. These analyses show a significant negative effect of Cloze, such that higher HA rates are associated with lower scores. However, we observe no effect of WMC, L1 proficiency, nor any interaction (Table 1).

Discussion. The slight trend towards HA shown by the population mean is potentially surprising for theories of L2 processing, as both Chinese and English are reportedly low-attachment (LA) languages. However, the bimodal distribution of attachment preferences provides support for recent arguments in favor of examining individual differences over population means. In contrast with Cotter and Ferreira (2024), we found no direct effect of WMC on attachment. This divergence might be explained by work arguing that L2 processing is by default more cognitively effortful than L1 processing. In this sense, high-span L2 users have been shown to pattern similarly to low-span L1 users (Dussias & Pinar 2010), possibly masking (or maxing out) any relevant differences in WMC effects during L2 (but not L1) disambiguation. More intriguingly for future work, the inverse effect of Cloze – a lexical proficiency measure – but not WMC directly is consistent with automaticity in lexical retrieval modulating L2ers' use of cognitive resources and choice of processing strategy (Bartek et al. 2011; Dekydtspotter et al. 2006; Hopp 2014; 2025).

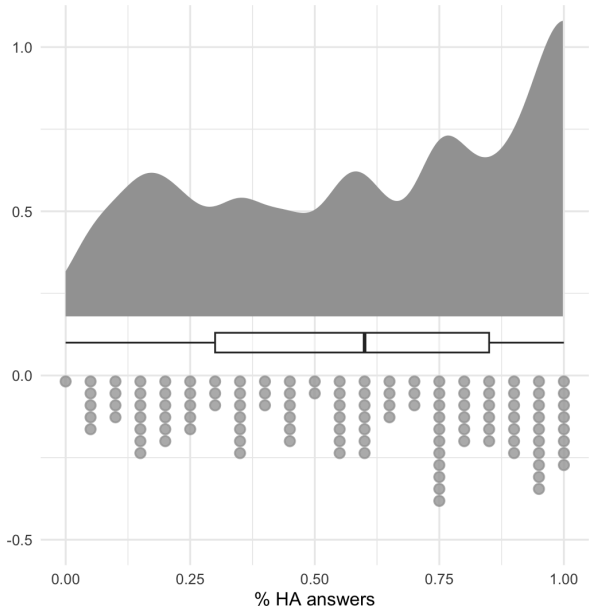

Figure 1: Distribution of HA preferences.

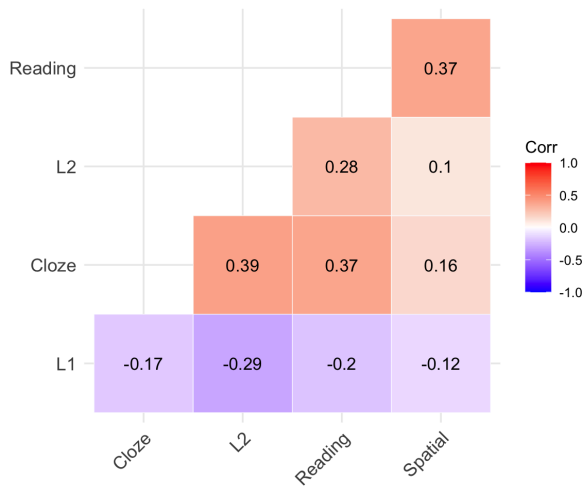

Figure 2. Correlation Matrix.

	Est.	Std. Error	z value	Pr(>|z|)
(Intercept)	0.50260	0.24334	2.065	0.0389 *
AVGM	-0.26564	0.25223	-1.053	0.2923
L1	0.07323	0.22594	0.324	0.7459
Cloze	-0.61544	0.21142	-2.911	0.0036 **
AVGM:L1	0.17054	0.32630	0.523	0.6012
AVGM:Cloze	0.03650	0.24735	0.148	0.8827
AVGM:L1:Cloze	-0.43923	0.34052	-1.290	0.1971

Table 2. Summary of the best fitting GLMER model.

Bartek et al. 2011. Journal of Experimental Psychology: Learning, Memory, and Cognition. **Cotter & Ferreira.** 2024. Memory & Cognition. **Dekydtspotter et al.** 2006. Proc. of GASLA. 2006. **Dussias & Piñar.** 2010. Second Language Research. **Hopp.** 2014. Language Acquisition. **Hopp.** 2025. Language Learning. **Li et al.** 2020. Bilingualism: Language and Cognition. **Montrul.** 2006. International Journal of Bilingualism. **Swets et al.** 2007. Memory & Cognition. **Omaki.** 2005. MA Thesis.